

■ Generative AI Bootcamp Structure

1. Introduction to Generative AI

- What is AI, ML, and Generative AI
- Traditional AI vs Generative AI
- Real-world GenAI use cases (ChatGPT, Copilot, Gemini, Midjourney)
- Types of GenAI: Text, Image, Audio, and Video

2. How Generative AI Works (Conceptual)

- The “predict the next word” intuition behind LLMs
- What are tokens and tokenization
- What are word embeddings
- What are Large Language Models (LLMs) — GPT, Llama, Claude, Gemini

3. Working with LangChain (Hands-On)

- What is an API and API keys in simple terms
- Overview of Langchain and components
- Basic RAG concept (Splitting, Chunking, Vector Databases)

4. Frontend and Backend

- Gradio UI setup
- Connecting backend (FastAPI) with GenAI API
- Handling user input and AI output
- Displaying results dynamically